Statistical Inference

Introduction and Hypothesis Testing

Vaibhavi Z

Introduction

- Statistical inference deals with **uncertainty in decision-making**.
- It uses **probability theory** to draw conclusions about a population.
- Conclusions are based on information obtained from a **sample**.
- The field has developed significantly since the **late 19th century**.

Meaning of Statistical Inference

- Statistical inference refers to making statements about **population parameters**.
- These statements are based on sample observations.
- The main aim is to infer population characteristics from sample data.

Problems in Statistical Inference

Statistical inference deals mainly with two types of problems:

1. **Hypothesis Testing**
    - Testing assumptions about the parent population.
2. **Estimation**
    - Using sample statistics to estimate unknown population parameters.

Hypothesis Testing

- Hypothesis testing begins with an assumption about a population parameter.
- A hypothesis is a **supposition made as a basis for reasoning**.

Definitions of Hypothesis

Morris Hamburg

"A hypothesis in statistics is simply a quantitative statement about a population."

Palmer O. Johnson

"Hypotheses are islands in the uncharted seas of thought, used as bases for consolidation and recuperation as we advance into the unknown."

Examples of Hypothesis

- A coin is tossed 200 times and results in 80 heads and 120 tails.
  - Test whether the coin is **unbiased**.
- Average weight of 100 students is found to be 110 lbs.
  - Test whether the population mean weight is **115 lbs**.
- Testing whether variables in a population are **uncorrelated**.

Procedure of Testing Hypothesis

The procedure of hypothesis testing involves the following steps:

1. Setting up the hypothesis
2. Collection of sample data
3. Calculation of sample statistics
4. Decision making

Setting up the Hypothesis

- Assume a value for a population parameter.
- Collect sample data and calculate the sample mean.
- Find the difference between hypothesised value and sample value.
- Smaller difference implies greater likelihood of correctness.

Types of Hypotheses

In practice, two mutually exclusive hypotheses are formulated:

- **Null Hypothesis (H_0)**
- **Alternative Hypothesis (H_1 or H_a)**

If one hypothesis is accepted, the other is rejected.

Rejection vs Acceptance of Hypothesis

- The rejection statement is much stronger than the acceptance statement.
- If the null hypothesis is not rejected, it does **not** mean it is proven true.
- In logic, it is easier to prove something **false** than to prove it **true**.
- Hence, statisticians avoid categorical acceptance of the null hypothesis.

Role of the Statistician

- Practical or managerial decisions are **outside the responsibility** of the statistician.
- The statistician does not make decisions.
- He only provides statistical information.
- This information assists businessmen or administrators in decision-making.

Errors in Testing of Hypothesis

When a statistical hypothesis is tested, four possible situations may arise:

1. Hypothesis is true but rejected **(Type I Error)**
2. Hypothesis is false but accepted **(Type II Error)**
3. Hypothesis is true and accepted **(Correct Decision)**
4. Hypothesis is false and rejected **(Correct Decision)**

Two Types of Errors

- Type I and Type II situations lead to **errors**.
- Correct decisions do not involve any error.

Type I Error

- A Type I error occurs when the null hypothesis is **rejected although it is true**.
- The probability of committing a Type I error is denoted by α (alpha).

$$\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Type II Error (β)

- Type II error occurs when a false null hypothesis is not rejected.
- The probability of Type II error is denoted by β.

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

Trade-off Between Type I and Type II Errors

- Due to a fixed sample size, it is not possible to control both types of errors simultaneously.
- There exists a **trade-off** between Type I error (α) and Type II error (β).
- Reducing the probability of one type of error
- Can only be achieved by increasing the probability of the other type of error.
- Therefore, careful selection of significance level is necessary.

Level of Significance

- It is more dangerous to accept a false hypothesis (Type II error) than to reject a true one (Type I error).
- Therefore, the probability of committing a Type I error is fixed in advance.
- This probability is called the **level of significance**.
- The level of significance is denoted by the Greek letter α.

Meaning of Level of Significance

- Level of significance is also known as:
  - Size of the test
  - Size of the rejection region
  - Size of the critical region
- It represents the probability of rejecting a true null hypothesis.
- In most statistical tests, α is fixed at 5%.

Interpretation of 5% Level of Significance

- If $\alpha = 0.05$, then:

$$P(\text{Rejecting a true } H_0) = 0.05$$

$$P(\text{Accepting a true } H_0) = 0.95$$

- Thus, the probability of accepting a true hypothesis is 95%.

One-tailed and Two-tailed Tests

- Hypothesis tests may be classified as:
    - One-tailed tests
    - Two-tailed tests
- The classification depends on the nature of the alternative hypothesis.

Two-tailed Test of Hypothesis

- In a two-tailed test, the null hypothesis is rejected if:
    - Sample statistic is significantly higher, or
    - Significantly lower than the hypothesised value.
- Rejection regions lie in **both tails** of the distribution.

Two-tailed Test at 5% Level of Significance

- Total significance level $\alpha = 0.05$.
- Rejection region in each tail:

$$\frac{0.05}{2} = 0.025$$

- Acceptance region on each side of the mean:

$$0.5 - 0.025 = 0.475$$

Critical Values for Two-tailed Test

- An area of 0.475 corresponds to 1.96 standard errors.
- Acceptance region lies between:

$$-1.96 \leq Z \leq +1.96$$

- If the sample mean lies within this range, H_0 is accepted.
- If it lies beyond this range, H_0 is rejected.

Reducing Type I Error

- To reduce the risk of committing a Type I error, the level of significance is reduced.
- A commonly used lower level is $\alpha = 0.01$.
- This means the probability of rejecting a true hypothesis is 1%.

Two-tailed Test at 1% Level of Significance

- Total significance level $\alpha = 0.01$.
- Rejection region in each tail:

$$\frac{0.01}{2} = 0.005$$

- Acceptance region on each side:

$$0.5 - 0.005 = 0.495$$

Critical Values at 1% Level

- An area of 0.495 corresponds to 2.58 standard errors.
- Acceptance region lies between:

$$-2.58 \leq Z \leq +2.58$$

- As α decreases, the acceptance region increases.

Effect of Decreasing α

- Decreasing the size of the rejection region reduces Type I error.
- However, it increases the probability of accepting the null hypothesis.
- When $\alpha = 0.01$, probability of rejecting a true hypothesis is 1%.

Two-tailed Test of Hypothesis

- A two-tailed test is appropriate when:

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu \neq \mu_0$$

- Here, μ_0 is a specified value of the population mean.
- The null hypothesis is rejected if the sample mean differs significantly
- Either above or below the hypothesised value.

Example of a Two-tailed Test

- Suppose we want to test:

$$H_0 : \mu = 50{,}000$$
$$H_1 : \mu \neq 50{,}000$$

- The hypothesis states that the average household income is 50,000.
- The null hypothesis will be rejected if:
    - Sample mean is too far above 50,000, or
    - Sample mean is too far below 50,000.
- Hence, the rejection region lies in **both tails**.

One-tailed Test of Hypothesis

- In a one-tailed test, the rejection region lies in only **one tail**.
- The location of the rejection region depends on the alternative hypothesis.
- One-tailed tests may be:
  - Right-tailed test
  - Left-tailed test

Right-tailed Test

- A right-tailed test is used when the alternative hypothesis is:

$$H_0 : \mu = 50{,}000$$

$$H_1 : \mu > 50{,}000$$

- All the α risk is placed in the **right tail**.
- The null hypothesis is rejected if the sample mean is significantly greater than 50,000.

Left-tailed Test

- A left-tailed test is used when the alternative hypothesis is:

$$H_0 : \mu = 50{,}000$$

$$H_1 : \mu < 50{,}000$$

- All the α risk is placed in the **left tail**.
- The null hypothesis is rejected if the sample mean is significantly less than 50,000.

Summary of One-tailed and Two-tailed Tests

Test Type	**Alternative Hypothesis**	**Rejection Region**
Two-tailed	$\mu \neq \mu_0$	Both tails
Right-tailed	$\mu > \mu_0$	Right tail
Left-tailed	$\mu < \mu_0$	Left tail

Universe Distribution

- A universe distribution arises when **each and every item** of the universe is studied.
- In this case, we have complete knowledge of:
  - Mean of the universe
  - Standard deviation of the universe
- The universe (true) mean is denoted by μ.
- The universe standard deviation is denoted by σ.

Parameters

- Greek letters are used to distinguish universe measures from sample measures.
- Measures that describe a universe are called **parameters**.
- Examples of parameters:
    - Population mean (μ)
    - Population standard deviation (σ)

Sample Distribution

- When only a part of the universe is studied, a **sample distribution** is obtained.
- The mean of the sample is denoted by $\bar{X}$.
- The standard deviation of the sample is denoted by S.
- Measures characterising a sample are called **statistics**.

Statistics

- A statistic is any numerical measure calculated from a sample.
- Examples:
  - Sample mean ($\bar{X}$)
  - Sample standard deviation (S)
- From a single universe, many different samples and sample distributions are possible.

Sampling Distribution

- A sampling distribution is the probability distribution of a statistic.
- It shows how a statistic behaves over repeated samples.

Features of Sampling Distribution

1. A sampling distribution is derived from a known or assumed population distribution.
2. The same population can generate infinitely many sampling distributions for different sample sizes.
3. A population may generate sampling distributions for different statistics.

Importance of Sampling Distribution

- Sampling distributions have well-defined properties.
- These properties help in:
    - Estimating population parameters
    - Calculating sampling errors
    - Making generalisations about populations

Sampling Distribution of Mean

The sampling distribution of the mean has the following properties:

1. The mean of the sampling distribution equals the population mean:

$$E(\bar{X}) = \mu$$

Standard Error of Mean

- The standard deviation of the sampling distribution of mean is called the **standard error**.

$$\text{S.E.}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

- Here, n is the sample size.

Normality of Sampling Distribution

- The sampling distribution of the mean is normally distributed.
- This property enables estimation of population mean using sample mean.

$$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

- This interval includes approximately 95% of all sample means.

Utility of Standard Error

The concept of standard error is important due to the following reasons:

1. Hypothesis testing
2. Measuring reliability of a sample
3. Determining confidence limits

Standard Error in Hypothesis Testing

- Hypotheses are usually tested at:
  - 5% level of significance
  - 1% level of significance
- If the difference between observed and expected is:
  - Less than 1.96 S.E. $\rightarrow$ Not significant
  - More than 2.58 S.E. $\rightarrow$ Significant at 1%

Three Standard Error Rule

- A difference greater than 3 S.E. is very unlikely due to chance.
- Probability of such a difference occurring by chance is only 0.27%.
- This rule is valid only for large samples.

Standard Error and Reliability

- Standard error indicates unreliability of a sample.
- Larger S.E. implies greater unreliability.
- Reliability (precision) is inversely related to S.E.
- To double precision, sample size must be increased four times.

Standard Error and Confidence Limits

- In a normal distribution:
    - Mean $\pm$ 1 S.E. covers 68.27%
    - Mean $\pm$ 2 S.E. covers 95.45%
    - Mean $\pm$ 3 S.E. covers 99.73%
- Probability of falling outside ± 3 S.E. is only 0.27%.

Estimation

- When data are collected by sampling, the main objective is to draw inferences about the population.
- Statistical estimation deals with methods of estimating population characteristics from sample data.
- The true value of a population parameter is an unknown constant.
- Complete enumeration of a population is often expensive or infeasible.

Need for Estimation

- In finite populations, complete censuses may be costly.
- In infinite populations, complete enumeration is impossible.
- Therefore, estimation aims to:
  - Obtain a guess of the true parameter value, or
  - Determine an interval of plausible values.
- Estimation also helps in assessing the accuracy of estimates.

Types of Estimates

There are two types of estimates:

1. Point Estimates
2. Interval Estimates

Point Estimates

- A point estimate is a single numerical value used to estimate a population parameter.
- It is obtained from a random sample of size n.
- The estimator is usually denoted by $\hat{\theta}$.
- Since the estimator depends on sample values, it is a random variable.

Example of Point Estimate

- Estimating the average income of a village as 8,875 is a point estimate.
- It gives a single best guess of the population parameter.

Interval Estimates

- An interval estimate gives a range of values within which the parameter is expected to lie.
- It is specified by:
  - Lower limit (L)
  - Upper limit (U)
- Such an interval is called a **confidence interval**.

Confidence Interval

- A confidence interval satisfies:

$$P(L \leq \theta \leq U) = 1 - \alpha$$

- L and U are called confidence limits.
- $1 - \alpha$ is the confidence coefficient.

Example of Interval Estimate

- If the average income is estimated to lie between 8,000 and 9,500:
- This is an interval estimate.
- It provides a range of plausible values for the parameter.

Point Estimate vs Interval Estimate

- Point estimates provide an exact numerical value.
- However, they do not indicate how close the estimate is to the true parameter.
- Interval estimates provide a range with a specified level of confidence.
- Hence, interval estimates are generally preferred in practice.

Estimate and Estimator

- An **estimate** is the numerical value obtained from a sample.
- An **estimator** is the statistical method used to obtain the estimate.
- Example:
  - Sample mean $\bar{X}$ is an estimator of population mean μ.

Properties of a Good Estimator

A good estimator should satisfy the following properties:

1. Unbiasedness
2. Consistency
3. Efficiency
4. Sufficiency

Unbiasedness

- An estimator is unbiased if its expected value equals the parameter.

$$E(\hat{\theta}) = \theta$$

- Some estimators are asymptotically unbiased.
- Bias is not always undesirable if it reduces overall error.

Consistency

- An estimator is consistent if it converges to the true parameter as sample size increases.
- As $n \to \infty$, the probability that the estimator differs from the parameter by a small amount approaches 1.
- Consistency is more important for large samples.

Efficiency

- Efficiency refers to the variance of an estimator.
- Among unbiased estimators, the one with smaller variance is more efficient.
- Smaller variance means estimates are more closely clustered around the true value.

Sufficiency

- An estimator is sufficient if it uses all information in the sample.
- If a sufficient estimator exists, no other estimator is required.
- It ensures complete utilisation of sample information.

Methods of Estimation

Two important methods of estimation are:

- Method of Least Squares
- Method of Maximum Likelihood

Tests of Significance

Based on data type and sample size, tests of significance are classified as:

1. Tests of Significance for Attributes
2. Tests of Significance for Variables (Large Samples)
3. Tests of Significance for Variables (Small Samples)

Attributes in Statistics

- Unlike variables, attributes are qualitative: presence or absence of a characteristic.
- Example: Literacy – people classified as **literate** or **illiterate**.
- Each sample unit can be classified as:
  - Success (A) – attribute present
  - Failure $(1 - A)$ – attribute absent
- Probability of success: p Probability of failure: $q = 1 - p$

Types of Tests for Attributes

1. Test for Number of Successes
2. Test for Proportion of Successes
3. Test for Difference between Proportions

Test for Number of Successes

- The number of successes in a sample follows a **Binomial Distribution**.
- Standard Error of Number of Successes:

$$S.E. = \sqrt{npq}$$

where

  - $n =$ sample size
  - $p =$ probability of success
  - $q = 1 - p =$ probability of failure

Illustration 1: Coin Toss

Problem: A coin was tossed 400 times, and heads appeared 216 times. Test the hypothesis that the coin is unbiased.
Solution:

- Null Hypothesis: Coin is unbiased, i.e., $p = 0.5$
- Expected number of heads:

$$E(\text{heads}) = n \cdot p = 400 \cdot 0.5 = 200$$

- Observed number of heads $= 216$
- Standard Error:

$$S.E. = \sqrt{npq} = \sqrt{400 \cdot 0.5 \cdot 0.5} = \sqrt{100} = 10$$

- Test Statistic (Z):

$$Z = \frac{\text{Observed - Expected}}{S.E.} = \frac{216 - 200}{10} = 1.6$$

Conclusion

- At 5% level of significance, the critical Z-value for a two-tailed test is 1.96.
- Since $Z = 1.6 < 1.96$, **fail to reject** the null hypothesis.
- Interpretation: There is no significant evidence to suggest the coin is biased.

  More examples